



DEPARTMENT OF COMPUTER ENGINEERING

FACULTY OF ENGINEERING UNIVERSITY OF RUHUNA

EC8207 – Applied Big Data Engineering

Pipeline Analysis Report

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EG/2020/3941

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1. Executive Summary

Report Period: 2026-02-25 22:13:00 – 2026-02-26 04:13:33

Generated: February 26, 2026 at 04:13:33

Schedule: Every 6 Hours (Airflow DAG)

Total Transactions: 3,463

Total Volume: \$ 10,027,055.21

Fraud Detected: 377 (10.89%)

Validated Clean: 3,086 (89.11%)

Reconciliation: BALANCED

2. Reconciliation Report

The reconciliation report ensures that every transaction entering the pipeline is properly accounted for. The fundamental equation is:

Total Ingress = Validated (Clean) + Fraud (Blocked)

$$\$10,027,055.21 = \$6,971,802.34 + \$3,055,252.87$$

Difference: \$0.00 — BALANCED

2.1. Transaction Volume Breakdown

Category	Count	Volume (USD)	% of Total Vol.
Total Ingress	3,463	\$ 10,027,055.21	100.00%
Validated (Clean)	3,086	\$ 6,971,802.34	69.53%
Fraud (Blocked)	377	\$ 3,055,252.87	30.47%

Table 1: Transaction volume reconciliation summary

2.2. Transaction Count: Clean vs. Fraud

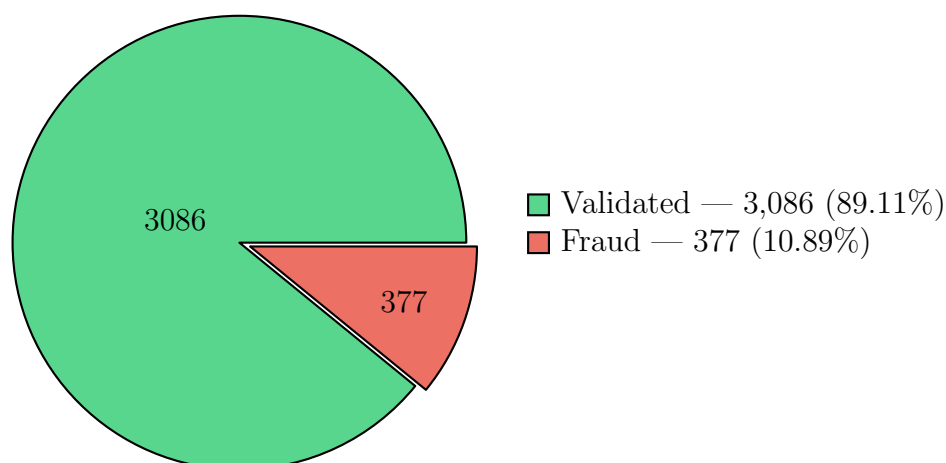


Figure 1: Transaction count distribution — Clean vs. Fraud

2.3. Volume Distribution: Clean vs. Fraud

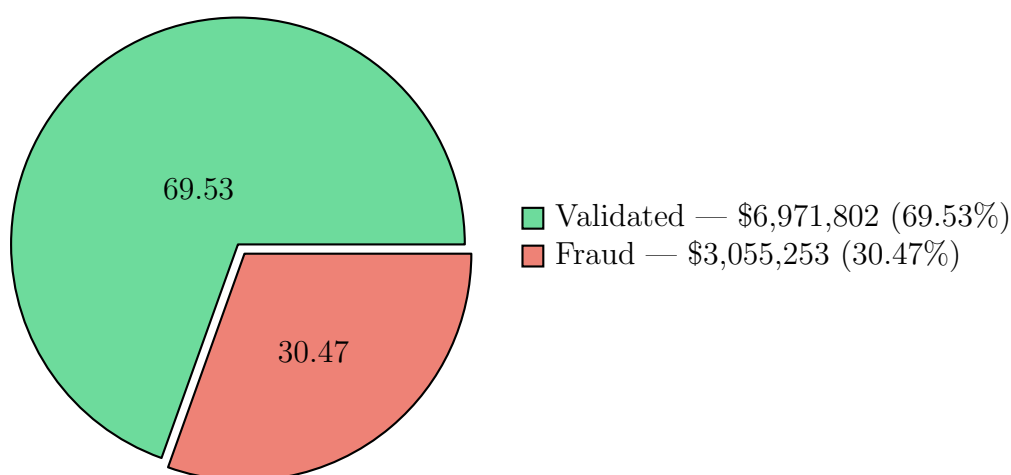


Figure 2: Transaction volume distribution — Clean vs. Fraud

Key Observation

While fraud accounts for only **10.89%** of transactions by count, it represents **30.47%** of total volume. This confirms that fraudulent transactions are heavily skewed toward high-value amounts, which is consistent with the **HIGH_VALUE** detection rule (transactions > \$5,000).

2.4. Average Transaction Value Comparison

The average validated transaction amount is calculated as: $\$6,971,802.34 \div 3,086 = \$2,259.69$. The average fraud transaction amount is: $\$3,055,252.87 \div 377 = \$8,104.12$.

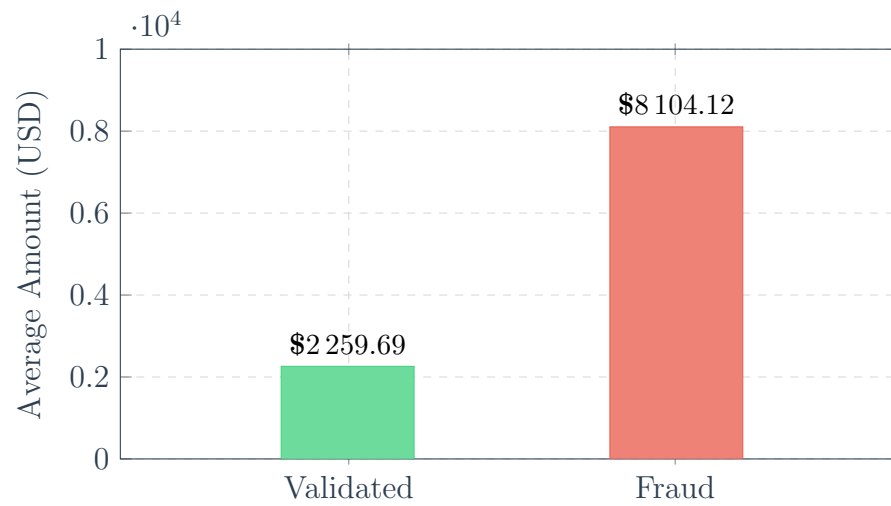


Figure 3: Average transaction value

The average fraud transaction (\$8,104.12) is **3.59x higher** than the average validated transaction (\$2,259.69). This makes sense because the **HIGH_VALUE** rule triggers on amounts exceeding \$5,000, pulling the fraud average upward significantly.

3. Fraud Analytics Report

3.1. Fraud by Detection Rule

The pipeline uses two detection rules. Based on the 377 flagged transactions:

Detection Rule	Count	Volume (USD)	% of Fraud
HIGH_VALUE (>\$5,000)	282	\$ 2,845,156.29	74.80%
IMPOSSIBLE_TRAVEL	95	\$ 210,096.58	25.20%
Total	377	\$ 3,055,252.87	100.00%

Table 2: Fraud breakdown by detection rule

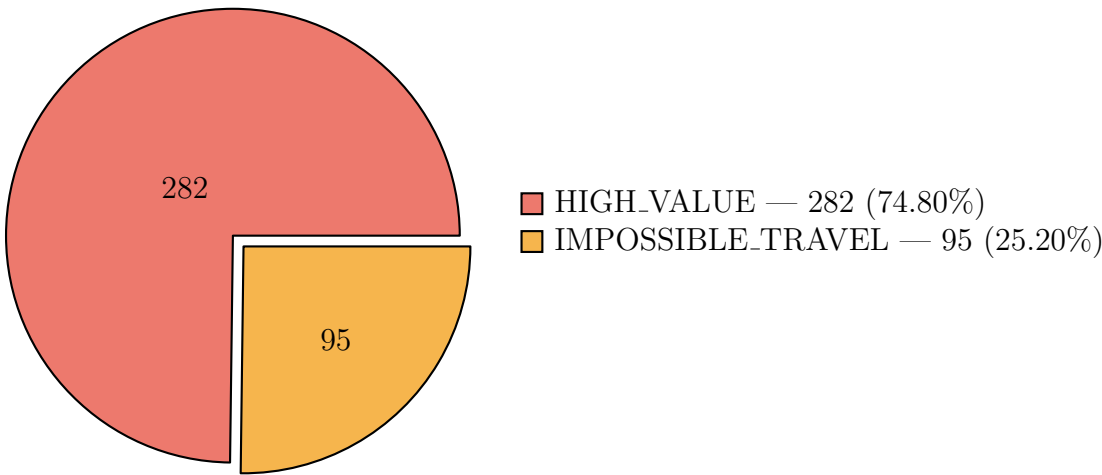


Figure 4: Fraud distribution by detection rule

3.2. Fraud Volume by Detection Rule



Figure 5: Fraud volume by detection rule — HIGH_VALUE dominates at \$2.85M

The `HIGH_VALUE` rule accounts for **93.12%** of fraud by dollar volume (\$2,845,156.29 ÷ \$3,055,252.87), even though it only accounts for 74.80% of fraud by count. The `IMPOSSIBLE_TRAVEL` rule catches 25.20% of fraud by count but only 6.88% by volume, since these are often normal-amount transactions from suspicious locations.

3.3. Fraud Attempts by Merchant Category

The table below shows how fraud is distributed across the 6 merchant categories:

Merchant Category	Count	% of Fraud	Volume (USD)	Avg (USD)
Food	75	19.89%	\$ 571,375.35	\$ 7,618.34
Healthcare	75	19.89%	\$ 593,671.37	\$ 7,915.62
Travel	60	15.92%	\$ 479,215.52	\$ 7,986.93
Utilities	58	15.38%	\$ 456,335.85	\$ 7,867.86
Fashion	55	14.59%	\$ 484,385.54	\$ 8,807.01
Electronics	54	14.32%	\$ 470,269.24	\$ 8,708.69
Total	377	100.00%	\$ 3,055,252.87	\$ 8,104.12

Table 3: Fraud attempts by merchant category

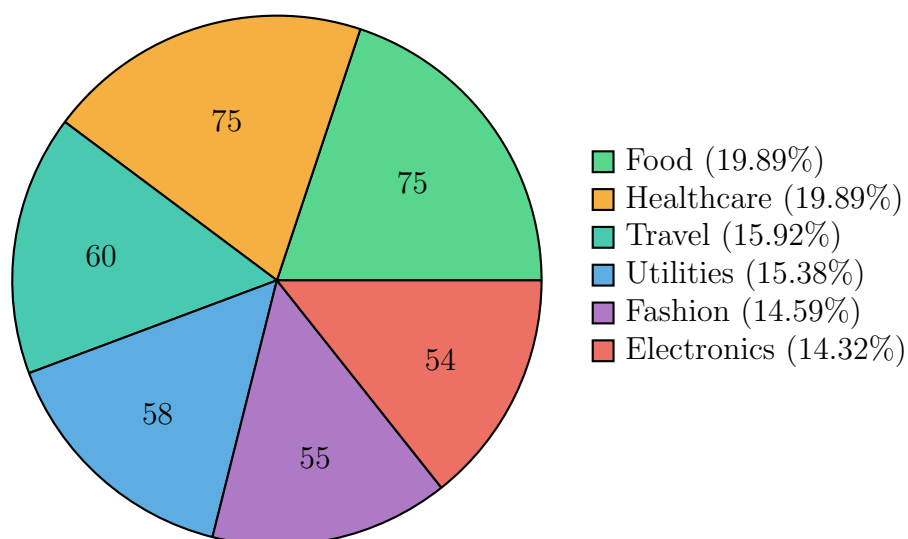


Figure 6: Fraud distribution by merchant category

3.4. Fraud Volume by Merchant Category - Bar Chart

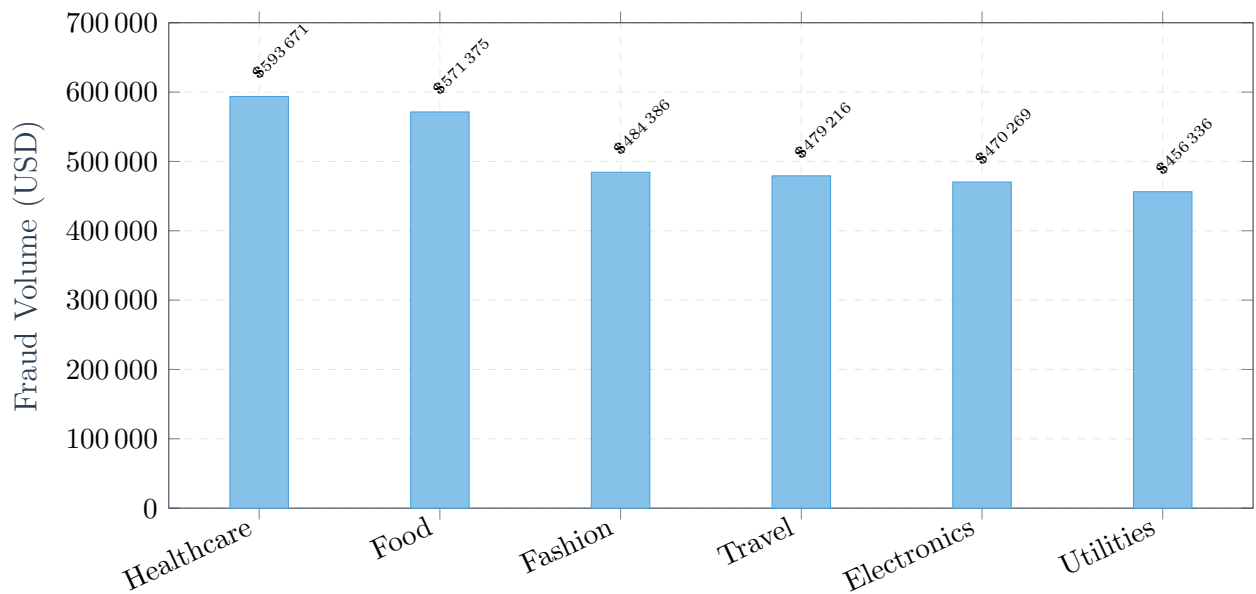


Figure 7: Fraud volume (USD) by merchant category, sorted by volume

3.5. Average Fraud Amount by Merchant Category

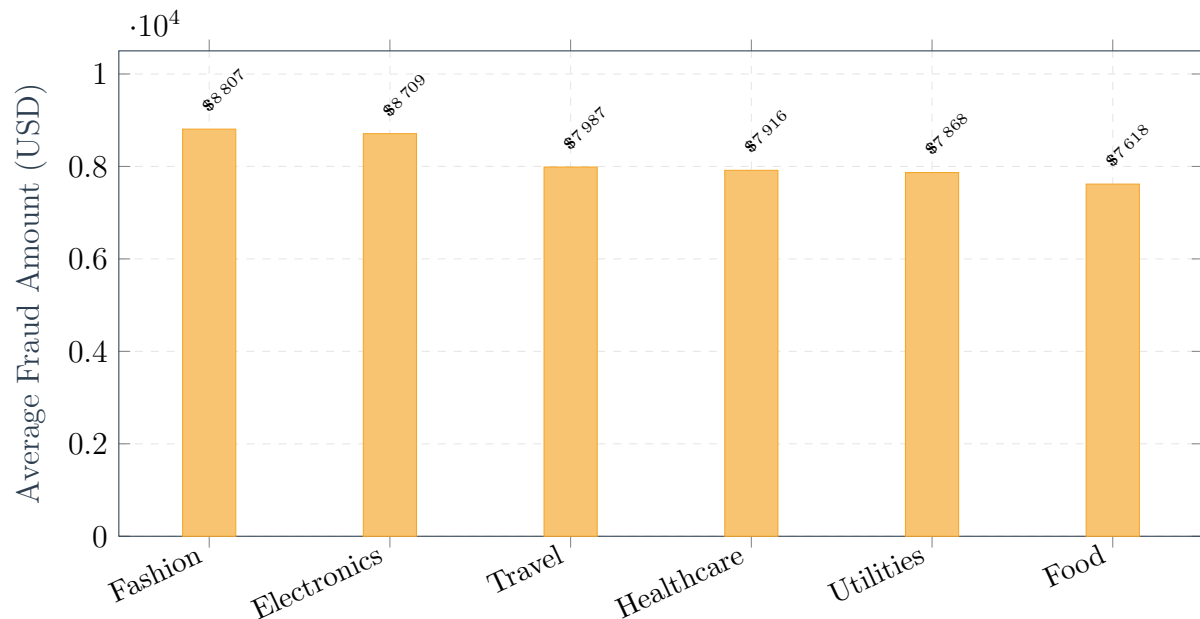


Figure 8: Average fraud amount by category

Category Insight

This period shows a more **even distribution** across all 6 categories. Food and Healthcare are tied at the top with 75 fraud attempts each (19.89%). **Fashion** (\$8,807.01) and **Electronics** (\$8,708.69) have the highest average fraud amounts per transaction, suggesting attackers target higher-priced items in these categories.

4. Fraud Rate and Risk Analysis

4.1. Key Risk Metrics

Metric	Value
<i>Overall Fraud Rates</i>	
Fraud Rate (by count)	10.89%
Fraud Rate (by volume)	30.47%
<i>Transaction Size Analysis</i>	
Average Fraud Transaction Size	\$ 8,104.12
Average Clean Transaction Size	\$ 2,259.69
Fraud-to-Clean Size Ratio	3.59×
<i>Risk by Merchant Category</i>	
Highest Risk Category (by count)	Food & Healthcare (tied)
Highest Risk Category (by avg. amount)	Fashion
Lowest Risk Category (by count)	Electronics
<i>Detection Rule Effectiveness</i>	
HIGH_VALUE Share (by count)	74.80%
HIGH_VALUE Share (by volume)	93.12%
IMPOSSIBLE_TRAVEL Share (by count)	25.20%
IMPOSSIBLE_TRAVEL Share (by volume)	6.88%

Table 4: Summary of key risk metrics

4.2. Fraud Rate vs. Volume Rate

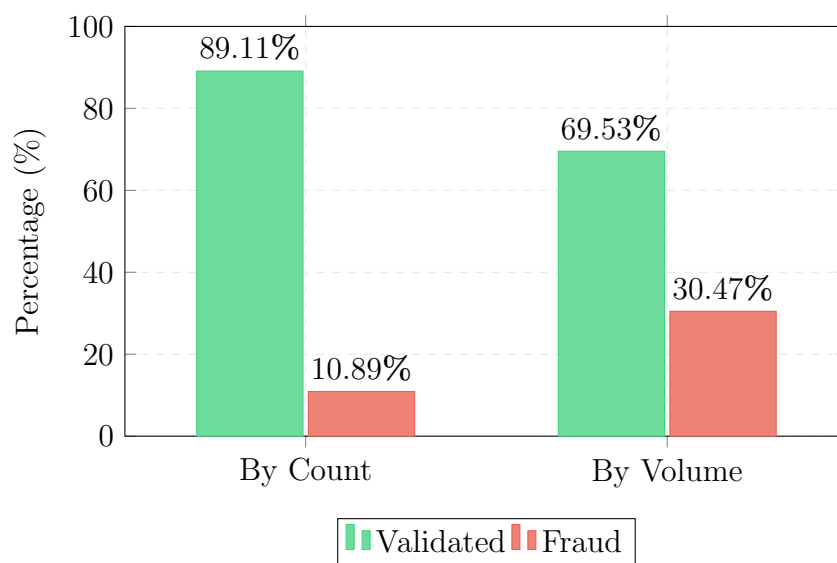


Figure 9: Fraud accounts for 10.89% of count but 30.47% of volume, high-value skew

Risk Insight

The gap between the fraud rate by count (10.89%) and by volume (30.47%) indicates that attackers specifically target high-value transactions. The average fraudulent transaction is worth \$8,104 compared to \$2,260 for legitimate ones. The `HIGH_VALUE` rule catches 74.80% of all fraud by count and a massive **93.12%** by dollar volume, making it the most critical detection rule in the pipeline. The `IMPOSSIBLE_TRAVEL` rule provides a complementary layer, catching 25.20% of fraud that might have smaller amounts but suspicious geographic patterns.

5. Pipeline Health Summary

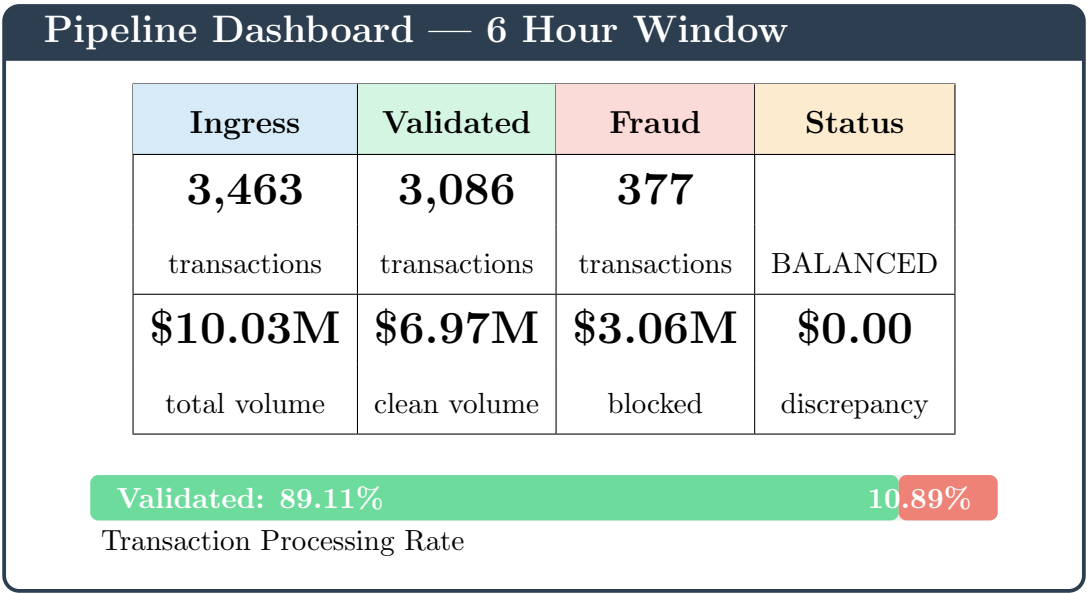


Figure 10: Pipeline health dashboard for the reporting window

6. Conclusions and Recommendations

Based on the analysis of 3,463 transactions processed in this 6-hour window:

- 1. Reconciliation is balanced.** Total ingress (\$10,027,055.21) exactly equals validated (\$6,971,802.34) plus fraud (\$3,055,252.87), confirming zero data loss in the pipeline.
- 2. Fraud distribution is evenly spread** across all 6 merchant categories this window, with Food and Healthcare tied at the top (75 each, 19.89%). No single category dominates, suggesting the attack pattern is broad rather than targeted at a specific merchant type.
- 3. Fashion and Electronics have the highest per-transaction fraud values** (\$8,807.01 and \$8,708.69 respectively). Even though they rank lower in count, the monetary impact per fraud is highest in these categories. Category-specific thresholds could help catch more nuanced attacks.
- 4. HIGH_VALUE rule is the primary detector**, catching 74.80% of all fraud by count and 93.12% by dollar volume. The IMPOSSIBLE_TRAVEL rule catches 25.20% of fraud by count but only 6.88% by volume, indicating these are often normal-amount transactions from geographically suspicious locations.
- 5. Volume disproportionality remains significant.** Fraud is 10.89% by count but 30.47% by volume, meaning each blocked fraud transaction saves the provider an average of \$8,104. The pipeline's real-time detection prevented an estimated **\$3.06 million** in potential losses during this 6-hour window.
- 6. Recommendation:** Consider adding a third detection rule for *velocity checks* e.g., more than 5 transactions within 2 minutes from the same user, to catch automated card testing attacks that may fall below the \$5,000 threshold. Additionally, category-specific thresholds (e.g., \$3,000 for Fashion/Electronics) could improve detection granularity.

End of Pipeline Analysis Report

Generated by Apache Airflow — Reconciliation DAG

Report Timestamp: 2026-02-26 04:13:33

Next scheduled run: 2026-02-26 10:13:33